

PGPilot — FINAL: Complete step-by-step System Modules for Tenant, PG Admin, and Super-Admin

Aftab bhai — yeh poora blueprint aap copy-paste karke implementation ya handoff de sakte ho. Har role ke liye **modules**, **DB fields (important)**, **API endpoints (key)**, **UI actions**, **step-by-step workflows**, **edge cases**, aur **qa/checklist** diya gaya hai. Har cheez sequence-wise (kis ke baad kya karna hai) bataaya gaya hai.

1 — Quick overview (one line)

Roles: **Tenant (user)**, **PG Admin / Manager (owner of a PG)**, **Super-Admin (you / SaaS owner — PGPilot)**.

Har role ke liye modules + stepwise actions niche hain.

2 — Shared primitives (DB + auth + common APIs)

Common DB tables used by all roles (already in system):

- pgs, rooms, beds, tenants, payments, mess_plans, tenant_mess, mess_bills, bills, bill_allocations, referral_codes, referrals, plans, add_ons, pg_subscriptions, billing_invoices, coupons, users, audit_logs.

Auth & roles

- users table: id, name, email, phone, password_hash, role
ENUM('superadmin', 'pg_admin', 'accountant', 'support', 'tenant'), pg_id (nullable), created_at
- JWT with role and pg_id claims.
- Password reset via OTP/email.

Common APIs

- POST /auth/login, POST /auth/logout, POST /auth/otp-login
- GET /me (returns role + permissions)

All steps below assume these primitives exist.

3 — TENANT ROLE (User app / PWA)

Goal: tenant can view rent, pay, see receipts, mess menu, raise complaints.

3.1 Features (module list)

1. Signup / OTP login

2. Profile & Documents (Aadhaar upload)
3. Assigned Bed & Room view
4. Due Rent (month view) + Pay Now (UPI/gateway)
5. Payment History & Receipt download
6. Mess / Food (if PG enabled) — menu, mess bill, mark meal taken (optional)
7. Complaints (maintenance)
8. Notifications (WA/email)
9. View subscription info? (only tenant-level receipts not needed)

3.2 Important DB fields (tenant)

- tenants: id, pg_id, bed_id, name, phone, email, aadhaar_url, join_date, rent_amount, deposit_amount, is_active

3.3 Key APIs (tenant)

- POST /auth/otp (login)
- GET /tenants/me
- GET /tenants/me/due → returns current due + month_for
- POST /payments/initiate → returns payment link
- POST /payments/webhook (gateway)
- GET /tenants/me/payments
- GET /mess/menu (if enabled)
- POST /complaints
- GET /payments/{id}/receipt

3.4 Tenant UI flow — Step by step (user onboarding → pay)

1. **Install / Open app** → Click “Login via OTP” → enter phone → OTP → authenticated.
2. GET /tenants/me → if profile missing, show “Complete Profile” form (name, aadhaar upload).
3. Show **Assigned Bed** (room name + bed #). If no bed assigned show “Contact PG Admin” or “Request Bed” (creates unassigned tenant request).
4. **Due Rent** card on home screen: Amount, Due Date, Pay Now button.
 - Click Pay Now → choose UPI/Gateway → POST /payments/initiate → redirect to gateway → on success webhook fires → show success screen + download receipt link + WA message sent.
1. **Mess tab** (if PG.food_enabled): view Today’s menu, Mess plan, Mess bill → Pay Mess bill same as rent or combined.
2. **Complaint**: simple form (title, desc, optional photo) → POST /complaints.
3. **Notifications**: WA/email for payment success, reminders, mess bill.

3.5 Edge cases (Tenant)

- Partial payments allowed → multiple payments per month; show aggregated paid amount.
- Tenant moves mid-month → prorate logic in rent view (if enabled).
- Tenant tries to pay for month already paid → show warning with last txn id.

3.6 QA / Acceptance checks (Tenant)

- OTP login works for 100% test phones.
- Payment link success and webhook reconciliation tested.
- Receipt PDF correct and downloadable.
- Mess bill generation matches meal logs.

4 — PG ADMIN / MANAGER ROLE (Main operator)

Goal: daily operations — add rooms/tenants, collect rent, manage mess, view pending list, share reminders.

4.1 Features (module list)

1. Dashboard (KPIs: total beds, occupied, vacant, monthly expected, received, pending)
2. PG settings (address, due day, food_enabled etc.)
3. Rooms & Beds management (create, edit, bed grid)
4. Tenant management (add, assign bed, checkout, upload documents)
5. Payments (mark cash, generate UPI link, gateway integration, receipts)
6. Pending Rent report (filter by month, send reminders)
7. WhatsApp reminder scheduler & templates
8. Mess / Food management (plans, assign, menus, meal logs, mess billing)
9. Bills (electricity) split and allocation
10. Referral dashboard (see your referrals/free days)
11. Subscription view (see plan, expiry, upgrade)
12. Support / tickets view

4.2 Important DB fields (PG Admin context)

- pgs: id, name, food_enabled, default_due_day
- rooms: id, pg_id, room_name, total_beds, rent_per_bed
- beds: id, room_id, bed_number, status
- payments and mess_bills as earlier

4.3 Key APIs (PG Admin)

- **Rooms/Beds:** GET /pgs/{pg}/rooms, POST /pgs/{pg}/rooms, GET /rooms/{id}/beds, POST /rooms/{id}/beds, PUT /beds/{id}
- **Tenants:** POST /tenants, GET /tenants?pg_id=, POST /tenants/{id}/assign-bed, POST /tenants/{id}/checkout
- **Payments:** POST /payments/create, GET /payments?pg_id=&month_for=, POST /payments/webhook
- **Reports:** GET /reports/pending-rents?pg_id=&month_for=
- **Whatsapp:** POST /whatsapp/send, POST /whatsapp/schedule
- **Mess:** POST /mess/plans, POST /tenants/{id}/mess/assign, POST /mess/bills/generate
- **Referral:** GET /referrals/list?pg_id=

- Subscription: GET /pgs/{pg}/subscription, POST /pgs/{pg}/subscribe

4.4 PG Admin UI flow — Step by step (day-to-day + special ops)

A. Initial PG setup (first time) — order:

1. **Create PG** (if self-signup) or Super-Admin creates PG.
2. **PG Settings**: set default_due_day, food_enabled, contact details.
3. **Add Rooms**: Add rooms one by one (room_name, total_beds, rent_per_bed).
4. **Create Beds**: Auto-generate bed rows for room (bed_number 1..N, status=vacant).
5. **Add First Tenant**: POST /tenants (name, phone, aadhaar) → Assign bed from vacant grid.
6. **Mark initial deposit & rent** (manual payment or online).
7. **Set WhatsApp templates & reminder schedule**.

B. Daily rent collection routine:

1. Admin opens **Pending Rent** for month.
2. For each pending tenant: call them or press **Send WA reminder** → WA API sends templated message.
3. If tenant pays cash: press **Mark Paid (cash)** → create payments record → generate receipt (PDF) → send WA.
4. If tenant pays online: webhook marks received automatically.
5. End of day: export CSV and reconcile with bank.

C. Tenant checkout:

1. Admin goes to tenant profile → Checkout → system asks settle dues and deposit refund calculation.
2. On confirm → tenants.is_active=0, tenants.bed_id=NULL, beds.status='vacant'.
3. Generate final settlement document (if needed).

D. Mess management (if enabled):

1. Create mess plans (monthly or per-meal).
2. Assign tenants to plan.
3. Optionally log meals (or assume default).
4. Generate mess bills monthly → attach to tenant account or include in invoice.
5. Collect mess payments via same payments module.

E. Vacancy handling:

1. Rooms page shows Vacancy summary.
2. When new walk-in arrives → quick flow: Add tenant → select vacant bed → assign → mark deposit.
3. If bed blocked for repairs → set bed status=blocked (excluded from vacancy).

4.5 Edge cases (PG Admin)

- Double assignment prevention: API should check bed.status == 'vacant' before assign.
- Partial payments and refunds: system must allow multiple payments with different month_for.

- Duplicate tenant detection: warn on same phone/Aadhaar.
- Mess plan change mid-month: prorate or start next month (configurable).

4.6 QA checklist (PG Admin)

- Assign/checkout updates bed status correctly.
- Pending Rent report accuracy (test with multiple payments).
- WA reminders send & template placeholders filled.
- Mess billing totals match meal logs or fixed plan.

5 — SUPER-ADMIN ROLE (PGPilot owner)

Goal: full platform control — plans, pricing, invoices, referral policies, user management, analytics.

5.1 Features (module list)

1. Plans management (create/modify plans & add-ons)
2. Coupons & promotions
3. Trials & referral configuration
4. Subscription management (per-PG override, grant free days, pause)
5. Invoices & billing (view, mark paid, refunds)
6. Onboarding approvals / manage PG clients
7. Analytics & MRR (revenue, churn, trial-to-paid)
8. Audit logs & billing_audit_logs
9. Support dashboard (tickets)
10. System settings (WhatsApp provider keys, payment gateway keys, backup settings)

5.2 Important DB fields (billing & superadmin)

- plans, add_ons, pg_subscriptions, billing_invoices, billing_audit_logs, coupons, referrals

5.3 Key APIs (Super-Admin)

- Plans: POST /api/v1/plans, PUT /api/v1/plans/{id}, GET /api/v1/plans
- Coupons: POST /api/v1/coupons, GET /api/v1/coupons
- Subscriptions: GET /api/v1/pgs/{pg}/subscription, POST /api/v1/pgs/{pg}/grant-free-days, POST /api/v1/pgs/{pg}/override-price
- Billing: GET /api/v1/invoices, POST /api/v1/invoices/{id}/mark-paid, POST /api/v1/invoices/generate
- Referral management: GET /api/v1/referrals, POST /api/v1/referrals/activate
- Admin users: POST /api/v1/users (create pg_admin or support)

5.4 Super-Admin UI flow — Step by step (setup → operations)

A. Initial platform setup:

1. Configure Payment Gateway (Razorpay keys), WhatsApp provider keys, S3 keys.
2. Create default Plans (Basic / Standard / Premium) and Add-ons.
3. Set global Trial days and Referral reward days.

4. Seed a few coupons for pilot.

B. Onboard a PG (manual / auto)

1. PG signs up → goes to trial (if auto) or manual approval required.
2. Super-Admin verifies PG (optional) → activate subscription or keep trial.
3. For paid activation, invoice generated and payment reconciled.

C. Manage subscriptions & billing

1. Use Subscriptions page to search PG → view subscription details.
2. Actions: Grant free days → system updates expiry_date and logs audit entry.
3. Override price → create custom_price and write audit log.
4. Generate invoices for renewals and one-time charges.

D. Monitor metrics

1. Dashboard shows MRR, Active PGs, trials expiring, referral conversions.
2. Drill into invoices & failed payments.

E. Support & Audit

1. View tickets, escalate.
2. Check billing_audit_logs for disputes.

5.5 Edge cases (Super-Admin)

- Coupon misuse / referral spam → implement server side checks: same phone/email, limit per IP, captchas.
- Tax/GST inclusion → include tax lines in invoices configurable per region.
- Refunds & chargebacks → create clear refund policy and record in audit logs.

5.6 QA checklist (Super-Admin)

- Invoice generation cron tested for thousands of PGs (queueing).
- Payment retries & webhook idempotency tested.
- Grant free days updates expiry correctly and logs audit.

6 — COMPLETE STEP-BY-STEP BUILD ORDER (what to implement in which sequence)

Follow this order to build MVP quickly and safely.

Phase 0 — Prep

1. Finalize schema (tables listed earlier).
2. Setup repo + CI pipeline + staging environment.
3. Provision infra: DB, S3, Redis, app servers.

Phase 1 — Core auth & PG basics (Week 1)

1. Implement Auth (OTP & password), roles & RBAC.
2. Implement `pgs` CRUD + PG settings UI.

Phase 2 — Rooms/Beds/Occupancy (Week 2)

1. Build Rooms & Beds CRUD; bed grid UI with color codes.
2. API for vacancy summary & occupancy %.

Phase 3 — Tenants & Assignment (Week 3)

1. Tenant CRUD, assign/checkout flows, unassigned tenant list.
2. UI flows for quick assign (vacant bed grid) and change bed.

Phase 4 — Payments & Pending (Week 4)

1. Payments table + manual mark paid + receipt generation (PDF).
2. Pending rent report API + admin page.

Phase 5 — Payment gateway + webhooks (Week 5)

1. Integrate Razorpay (or chosen gateway) + webhooks + idempotency handling.
2. Tenant pay flow & admin reconciliation.

Phase 6 — WhatsApp reminders & scheduler (Week 6)

1. Setup WhatsApp provider + template editor + job queue (BullMQ).
2. Implement manual & scheduled reminders.

Phase 7 — Mess module & bills (Week 7)

1. Mess plans + assign + meal logs + mess billing.
2. Electricity bill split module.

Phase 8 — Referral & Subscription (Week 8)

1. Referral codes + activation flow.
2. Plans/add-ons + subscription table + invoices + trial logic.

Phase 9 — Super-Admin billing & analytics (Week 9)

1. Super-Admin pages: Plans, Add-ons, Coupons, Subscriptions, Invoices.
2. Billing cron & invoice generation worker.

Phase 10 — QA, E2E, Launch (Week 10)

1. E2E tests (tenant pay flow, admin mark paid, checkout).
2. Beta onboarding 5–10 PGs.
3. Fix issues, prepare marketing kit, launch.

7 — OPERATIONS & RUNBOOK (for admin / support)

- Daily: check failed payments, reconcile manual payments, respond to support tickets.
- Weekly: backup verification, review pending rents report, SMS/WA quota monitoring.
- Monthly: run invoices export & accountant sync, review MRR metrics.
- Incident: if payment gateway down → pause online payments and enable manual flows; notify customers.

8 — TEST CASES (must cover)

- Bed assign then checkout: bed status updates.
- Multiple payments for same month: paid aggregated.
- Mess plan assigned mid-month: prorated or full next month.
- Referral reward applied only after activation rule satisfied.
- Coupon limits (max_uses) respected.
- Idempotent webhook handling (duplicate webhooks don't double mark).

9 — SAMPLE API & SQL snippets (quick references)

Assign bed (API)

```
POST /tenants/{id}/assign-bed
```

```
Body: { "bed_id": 12 }
```

Server logic:

```
- BEGIN TRANSACTION
- SELECT status FROM beds WHERE id=12 FOR UPDATE
- if status!='vacant' rollback & return error
- UPDATE beds SET status='occupied' WHERE id=12
- UPDATE tenants SET bed_id=12 WHERE id={id}
- COMMIT
```

Pending rents SQL

```
SELECT t.id, t.name, COALESCE(t.rent_amount, r.rent_per_bed) AS rent_amount,
COALESCE(SUM(p.amount),0) AS paid
FROM tenants t
```



```
LEFT JOIN beds b ON b.id = t.bed_id

LEFT JOIN rooms r ON r.id = b.room_id

LEFT JOIN payments p ON p.tenant_id = t.id AND p.month_for = '2025-11-01' AND
p.status='received'

WHERE t.pg_id = 123 AND t.is_active = 1

GROUP BY t.id

HAVING paid < rent_amount;
```

10 — Final Checklists (copy & use)

Launch-ready checklist (PG Admin)

- Add PG contact & settings
- Add rooms + beds
- Add first 3 tenants & assign beds
- Set WhatsApp templates
- Mark first payment & checkout receipt delivery
- Run pending rents report

Launch-ready checklist (Super-Admin)

- Configure payment & WA provider keys
- Create default plans & add-ons
- Set trial days & referral reward
- Seed coupon for pilot
- Confirm backup cron & monitoring

Aftab bhai — maine pura system **step-by-step** diya hai (Tenant, PG Admin, Super-Admin) with implementation order.

Agar chaho to ab main **kisi ek** deliverable seed kar doon (choose one):

- A) Full SQL migration files (all tables)
- B) OpenAPI (Swagger) spec for all endpoints
- C) React Admin skeleton (pages & components)
- D) Payment & billing worker sample code (Prorate, Free days)
- E) WhatsApp templates set + demo messages in Hindi

Kaunsa chahoge pehle — main turant bana deta hoon.